Freshness Is Consumer-Relative and Witnessed: Sealed Measurement of Adaptation-Certificate Vacuity in 5G/6G Radio Access

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Abstract

Radio-access networks run on *adaptation certificates*: a channel-quality indicator (CQI) that certifies a modulation and coding scheme, a beam report that certifies a direction, a channel-state-information (CSI) feedback that certifies a precoder, a reference-signal-received-power (RSRP) report that certifies a serving cell. Each is a claim that a transmission decision is supportable. We argue, and measure, that such certificates are *vacuous under aging*: when the certificate ages faster than its report cadence the certified decision fails, yet the certificate still reads "adequate." We formalize the resulting *false-clear rate* as a consumer-relative quantity graded against an independent *witness*—the hybrid-ARQ (HARQ) acknowledgement, or the served-link outcome—and we measure it under a sealed-preregistration protocol: bars committed before any graded run, a structural cooling-off enforced in code, manipulation checks that are themselves bars, and graded seeds disjoint from calibration. Across four cells on a real 3GPP 5G New Radio (NR) physical layer (5G low-density parity-check coding and TR38.901 channels), the naive certificate false-clears at 2.6–$4.4\times$ the 10% target block-error rate, while a witnessed policy holds at target on every graded seed: CQI aging (0.34–0.37 vs. outer-loop 0.10), millimeter-wave beam aging (0.31–0.33 vs. beam-failure recovery 0.10), learned CSI-feedback reconstruction vacuity (0.27–0.30 vs. consumer-aware 0.13), and stale-RSRP handover (0.26–0.34 vs. radio-link-monitoring re-selection 0.10). We derive and measure a *refresh-floor law*—the maximum report period holding false-clear at target is proportional to the channel coherence time (measured slope 0.177, $R^2 = 0.915$). The CSI-feedback cell is the compression-domain instance: a codec optimal in reconstruction error still false-clears at the consumer because the precoder reads a low-dimensional projection of the channel, not its Frobenius norm.

Index Terms

Link adaptation, CSI feedback, beam management, handover, hybrid ARQ, age of information, reliability, pre-registration, 5G NR, 6G.

I. INTRODUCTION

A modern radio link is a chain of decisions taken on the basis of *reports*. The user equipment (UE) measures the channel and reports a CQI; the base station (gNB) selects a modulation and coding scheme (MCS) to hit a target block-error rate (BLER), conventionally 10%. The UE reports a best-beam index; the gNB steers a narrow beam to it. The UE feeds back compressed CSI; the gNB computes a precoder from it. The UE reports the RSRP of neighboring cells; the network decides which cell serves. Each report is, in effect, a *certificate*: an assertion that the decision it induces is supportable.

Certificates are trusted at a cadence. Between reports the channel evolves. When the channel decorrelates faster than the report period—high Doppler, fast rotation, mobility—the certificate *ages*: the decision it certified is no longer supportable by the time the transmission occurs, and it fails. The failure is observable to the network only through an independent channel: the HARQ acknowledgement (ACK/NACK), which reports whether the receiver actually decoded. We call a report that certifies "adequate" while the witnessed transmission fails a *false-clear*, and its rate the *false-clear rate*. This paper measures the false-clear rate of the four principal 5G/6G adaptation certificates and shows, in each case, that (i) the naive fixed-cadence certificate is measurably vacuous—its false-clear rate is far above the target BLER—and (ii) a policy that grades the certificate against the HARQ witness holds the false-clear rate at target.

A. Freshness is not a property of the report

The central conceptual claim is that freshness—whether a certificate is "fresh enough"—is not a property of the report in isolation. It is a *relation* among the report, the consumer that acts on it, and the channel dynamics that age it. Two consumers reading the same report at the same instant can experience different false-clear rates because they read the channel through different functionals: a robust low-rate link tolerates a stale CQI that a high-rate link cannot; a wide beam tolerates an angular drift that a pencil beam cannot. This *consumer-relativity* is the radio-access instance of a thesis we develop generally elsewhere [1]: operational reliability is a property of the observer relation, not of the observed object. Age-of-information

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Every empirical claim in this paper is governed by a sealed pre-registration and a coded verdict runner committed before the graded runs; the pre-registrations, runners, and graded records are archived with the Observation-Theory program. The physical-layer substrate is the open-source Sionna library over an NVIDIA Quadro GV100.

theory [2] quantifies staleness on the report channel; consumer-relative false-clear quantifies its *consequence* at the receiver, which is what a reliability claim must ultimately bound.

B. Contributions

1) A unifying framework (Section II): every 5G/6G adaptation certificate has an independent witness (HARQ or the served-link outcome), a consumer-relative false-clear rate, and a derivable refresh floor.
2) A sealed-preregistration measurement protocol (Section III) that makes each reliability claim falsifiable before it is measured, adapted from the Observation-Theory program to radio-access experiments.
3) Four sealed cells on a real 5G NR physical layer (Sections IV–V): CQI aging, beam aging, learned CSI-feedback vacuity, and stale-RSRP handover. Every bar and manipulation check passes on three graded seeds disjoint from calibration.
4) A measured refresh-floor law (Section VI): the maximum report period holding false-clear at target scales linearly with coherence time.
5) The CSI-feedback cell (Section V-C) demonstrates a *reconstruction paradox*: a codec that wins the reconstruction metric it was trained on loses at the consumer, the compression-domain instance of the same grammar.

We claim measurement discipline and a cross-cutting framework, not new adaptation mechanisms; outer-loop link adaptation, beam-failure recovery, and radio-link monitoring are credited prior art. The delta is the *measured, consumer-relative, calibrated* false-clear rate and the witnessed alternative, reported under a protocol designed to make the claim refutable.

II. Certificates, Consumers, and Witnesses

Definition 1 (Adaptation certificate)**.** *A report r_t emitted at time t that induces a network decision $a_t = \pi(r_t)$ (an MCS, a beam, a precoder, a serving cell) intended to satisfy a performance target on the resulting transmission.*

Definition 2 (Witness)**.** *An independent binary outcome $w_t \in \{ACK, NACK\}$ of the transmission taken under a_t, observed after the fact. For all four cells the witness is HARQ or, equivalently at the mobility layer, whether the served link decoded.*

Definition 3 (False-clear rate)**.** $\mathrm{FC}(\pi) = \Pr[w_t = \mathit{NACK} \mid a_t = \pi(r_t)]$, *the probability that the certified decision fails at the witness. A monitor is* vacuous *when* FC *exceeds the target it purports to certify.*

Freshness enters through the aging of r_t. Let τ be the report period and T_c the channel coherence time. The certificate is measured at the last report instant and applied until the next; the mismatch grows with τ/T_c. A *witnessed policy* augments π with feedback from w_t: outer-loop link adaptation (OLLA) adds a HARQ-driven offset to the CQI; beam-failure recovery (BFR) re-selects on consecutive NACKs; radio-link monitoring (RLM) triggers re-selection of the serving cell. Each converts the open-loop certificate into a closed loop around the witness.

Proposition 1 (Refresh floor, informal)**.** *For a fixed target β and operating SINR, there is a maximum report period $\tau^\star(\beta)$—the* refresh floor*—below which the naive false-clear stays at or under β, and $\tau^\star$ is proportional to the coherence time T_c of the aging process.*

Proposition 1 is the radio-access form of a refresh-floor law we state generally in the program; we measure its constant in Section VI. Intuitively, the certificate must be refreshed on the timescale over which the channel it certifies decorrelates.

III. Sealed-Preregistration Measurement

Reliability claims are unusually easy to overstate: thresholds, seeds, and success criteria chosen after seeing results manufacture significance. We therefore bind each cell to a protocol that fixes the claim before the graded run.

Bars. Each cell commits three bars—B1 (the naive certificate is vacuous), B2 (the witnessed policy holds), B3 (dominance: the witnessed false-clear is at most half the naive)—as numeric thresholds in a pre-registration document, together with three *manipulation checks* (MCs) that are also bars: MC1 that the aging is real (a nonzero certificate error), MC2 that the control with a fresh certificate is sane (so failures are attributable to aging, not a broken link), and MC3 non-degeneracy (the certificate is actually exercised). *Any* MC failure voids the cell.

Cooling-off. A verdict runner refuses to grade unless the pre-registration is sealed at least one day after its construction date; the rule is enforced in code, not convention. *Disjoint seeds.* Bars are calibrated on shakedown seeds $\{0, 1, 2\}$; the sealed verdict is computed on graded seeds $\{s_1, s_2, s_3\}$ never seen during calibration. *Kept failures.* A failing graded run is recorded as executed; the protocol has no escape hatch. The four cells below were constructed on one day and sealed and graded on the next, on graded seeds {20260823, 20260824, 20260825} disjoint from calibration.

TABLE I
SEALED CELLS, GRADED ON DISJOINT SEEDS $\{20260823–25\}$ ON THE REAL 5G NR SUBSTRATE. RANGES SPAN THE THREE GRADED SEEDS. EVERY BAR (B1–B3) AND MANIPULATION CHECK (MC1–MC3) PASSED ON EVERY SEED.

Cell	Certificate → decision	Witness	naive FC	witnessed FC	control	verdict
CSI (§V-A)	CQI → MCS	HARQ ACK/NACK	0.34–0.37	OLLA 0.10	fresh 0.11	PASS
BEAM (§V-B)	beam index → beam	HARQ / BFR	0.31–0.33	BFR 0.10	fresh 0.02	PASS
AICSI (§V-C)	CSI feedback → precoder	HARQ	0.27–0.30	aware 0.13	recon paradox[1]	PASS
HO (§V-D)	RSRP → serving cell	served-link	0.26–0.34	RLM 0.10–0.11	fresh 0.08–0.09	PASS

IV. Substrate: A Real 5G NR Physical Layer

All decoding is performed on standard-compliant 5G NR physical-layer components from the open-source Sionna library [3]: 5G LDPC encoding and decoding [4] with 20 belief-propagation iterations, Gray-mapped M-QAM, and TR38.901 tapped-delay-line channels [5]. For each MCS in a 13-entry span of the 3GPP MCS table [6] (rate-floored at $1/5$, the LDPC minimum), we measure a real BLER-versus-SINR curve by batched Monte-Carlo decoding on a GPU; these curves are the link-level model of the actual NR decoder, not a fitted sigmoid. The time-varying channel is a real TR38.901 process; per-transmission HARQ outcomes are drawn from the measured curve at the *actual, aged* SINR—the standard link-to-system methodology, with real link curves. This is one rung below a full system-level stack and several rungs above a synthetic fading simulator; we state the ladder explicitly and label each cell by the rung it occupies.

V. Four Sealed Cells

Table I summarizes the sealed, graded results. All cells target $\beta = 10\%$ BLER. We describe each in turn.

A. CSI aging: CQI → MCS

The UE reports a CQI every $\tau = 20$ transmission-time intervals; the gNB selects the highest MCS whose measured BLER-at-target SINR is supportable. The channel is TR38.901 TDL-A at Doppler $f_d = 200\,\text{Hz}$ ($v \approx 17\,\text{m/s}$ at $3.5\,\text{GHz}$), coherence time $T_c = 0.423/f_d \approx 2.1\,\text{ms}$, so the report is stale by roughly ten coherence times. Three policies share each channel realization: *naive* (raw CQI), *OLLA* (a HARQ-driven SINR offset with equilibrium at target), and *fresh* (a per-interval CQI, the no-aging control). Graded, the raw CQI false-clears at 0.34–0.37—3.4–3.7× the 10% target—while OLLA holds at 0.10 and fresh at 0.11; the mean reported-versus-realized SINR error is 5.6–5.9 dB (MC1), confirming a genuine aging regime.

B. Beam aging: index → beam

At millimeter wave the gNB serves through a narrow analog beam chosen from the UE's reported best-beam index. We model the beamforming gain with an exact 16-element half-wavelength uniform-linear-array factor and drive the true angle at $300°/\text{s}$; the block decode uses the same real NR LDPC curves. The reported beam is measured with $1°$ error every 20 slots and held. *naive* holds the periodic beam; *BFR* re-selects on two consecutive NACKs (the deployed witnessed correction [7]); *fresh* re-selects every slot. Graded, the stale beam false-clears at 0.31–0.33, BFR holds at 0.10, fresh at 0.02; the mean beamforming-gain loss under the naive policy is 4.7 dB (MC1). The angular refresh floor is the beam-domain form of Proposition 1.

C. Learned CSI feedback: the reconstruction paradox

In (6G) AI-native CSI feedback the UE compresses the measured MIMO channel with a learned codec; the gNB reconstructs it and derives a precoder. Codecs are conventionally trained to minimize reconstruction error (normalized mean-squared error, NMSE) [8]. But the consumer—the precoder and the served link—reads the channel through its dominant eigen-direction, a low-dimensional projection, not the Frobenius norm. We train two autoencoders with an identical architecture and bottleneck on 2×16 channels with angular structure: one on reconstruction NMSE, one on the achieved beamforming gain of the induced maximum-ratio precoder (a tail-sensitive, log-gain objective). The consumer's SINR is the operating SINR plus the achieved gain fraction in dB; HARQ is drawn from the real NR curves.

Graded, the reconstruction-optimal codec attains *lower* NMSE (0.66–0.68) than the consumer-aware codec (1.13–1.19)—it wins the metric it was trained on—yet false-clears at the consumer at 0.27–0.30, while the consumer-aware codec holds at 0.13. A perfect-CSI control false-clears at 0.0, so the link is sane; the aware codec's achieved gain exceeds the NMSE codec's on every seed (the mechanism). This is the compression-domain instance of the same grammar: *reconstruction fidelity is a certificate that false-clears at the consumer*, exactly the consumer-relative distortion principle we ship publicly in the compression setting [9]. It argues that AI-native CSI codecs should be trained and *certified* against the HARQ witness, not reconstruction NMSE.

[1]The reconstruction-optimal codec attains *lower* normalized reconstruction error (0.66–0.68) than the consumer-aware codec (1.13–1.19) yet a *higher* consumer false-clear—the paradox is the result (Section V-C).

D. Handover under stale RSRP

The UE's RSRP measurement report certifies a serving cell. Under mobility the report ages: the UE crosses the cell edge but the stale periodic report keeps it on the old cell, whose RSRP has dropped and whose former neighbor now interferes, collapsing the serving SINR. We use a two-cell interference-limited geometry (pathloss, correlated shadowing L3-filtered for handover decisions, a mild fast-fade term for the PHY) with the serving-link BLER from the real NR curves; a UE at $30\,\mathrm{m/s}$ oscillates across the edge with a 300-slot report period. *naive* hands over only on the stale report; *RLM* re-measures and re-selects on three consecutive serving NACKs (the outcome-witnessed correction); *fresh* tracks the best cell. Graded, the stale report false-clears at 0.26–0.34, RLM holds at 0.10–0.11, fresh at 0.08–0.09; the naive policy spends 24–31% of the time on a non-best cell (MC1). This is the radio-access twin of a routing-under-staleness problem—which replica, or cell, to serve from when the freshness certificate is stale—and is best graduated to a system-level multi-cell simulator, which we note as external-validity work.

VI. The Refresh-Floor Law, Measured

Proposition 1 predicts the refresh floor scales with coherence time. We measure it on the CSI substrate by sweeping Doppler $f_d \in \{10, 25, 50, 100, 200, 400\}$ Hz against report period, recording the largest period holding the naive false-clear at or under $1.5\times$ target. The floor falls from $7.4\,\mathrm{ms}$ at $f_d = 10\,\mathrm{Hz}$ ($T_c = 42\,\mathrm{ms}$) to $1.2\,\mathrm{ms}$ at $f_d = 400\,\mathrm{Hz}$ ($T_c = 1.1\,\mathrm{ms}$). A regression through the origin gives

$$\tau^\star \approx 0.177\, T_c, \qquad R^2 = 0.915,$$

i.e., the report must be refreshed roughly every one-sixth of a coherence time at this SINR and target. The high-Doppler points sit slightly above the line at the one-interval reporting-rate wall (the report cannot be faster than every slot), which we disclose rather than trim. The law converts a qualitative "report often enough" into a quantitative floor derivable from the UE's own Doppler.

VII. Discussion and Related Work

Adaptation mechanisms. OLLA [10], BFR, and RLM are deployed, HARQ- or outcome-witnessed corrections; we credit them and measure what they buy. The contribution is not a new controller but the calibrated false-clear rate of the naive certificate and the demonstration that a single grammar—certificate, witness, refresh floor—spans CQI, beam, CSI feedback, and handover.

AI-native CSI. Learned CSI feedback is typically evaluated by reconstruction NMSE or cosine similarity [8]. The reconstruction paradox (Section V-C) shows that metric can be won while the consumer loses, and argues for consumer-aware training and HARQ-witnessed certification.

Staleness and age of information. Age-of-information [2] bounds staleness on the report channel; consumer-relative false-clear measures its consequence at the receiver, the quantity a reliability target constrains. The same certificate/witness grammar recurs in distributed data systems, where replication-freshness monitors are consumer-relative vacuous and a write-ahead-log–witnessed certificate holds [1]; probabilistically bounded staleness [11] and commit-wait [12] are the database analogues of the refresh floor and the witness.

VIII. Limitations and External Validity

The substrate is link-level: real NR LDPC decoding and real TR38.901 channels, but a link-to-system HARQ draw rather than full per-block per-slot decoding, and narrowband SISO for the CQI and beam cells. The learned-CSI cell uses a continuous bottleneck as a feedback-bit proxy and an MRT precoder; the handover cell is a two-cell link proxy. Each cell names its rung and its graduation: full per-block decoding, wideband MIMO, a quantized CsiNet-class codec, and a system-level multi-cell simulator with time-to-trigger and radio-link-failure timers. The bars are cell-specific and calibrated with margin; we report the demonstrated values alongside the committed thresholds so the margins are transparent. Normative air-interface behavior is 3GPP's remit; this paper is measurement methodology.

IX. Conclusion

Freshness in 5G/6G radio access is not a property of a report but of the relation among the report, the consumer, and the channel. Measured under a protocol designed to make each claim refutable, the four principal adaptation certificates—CQI, beam, CSI feedback, RSRP—are each vacuous under aging at 2.6–$4.4\times$ their target, and each is repaired by grading against the HARQ or served-link witness. The maximum safe report period scales with coherence time (slope 0.177). The learned-CSI cell shows the same failure in the compression domain: reconstruction fidelity is a certificate that false-clears at the consumer. We recommend that adaptation certificates—and, increasingly, learned ones—be specified, measured, and *certified* against the witness that reports what the consumer actually received.

Reproducibility

Each cell's pre-registration, coded verdict runner (with the cooling-off and substrate guards), family record, and graded verdict are archived with the program [13]. The physical-layer curves are regenerable from the open-source substrate [3].

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